

Ashish Singh

Backend Developer & Distributed Systems

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PROFESSIONAL SUMMARY

Backend Developer specializing in **automated, smooth-running solutions** via AI and Distributed Architectures. Proficient in **Java, Python** with expertise in **Agentic AI orchestration**, RAG pipelines, and high-concurrency systems. Dedicated to engineering robust software that meets strict **regulatory and industry standards**.

EDUCATION

Vellore Institute of Technology
Integrated M.Tech in Software Engineering

Vellore, Tamil Nadu
Sep 2022 – July 2027

Class XII: 92.4% | Class X: 94.2%

Relevant Coursework: DSA, DBMS, OS, Computer Networks

TECHNICAL SKILLS

Languages & Frameworks: Java (21+), Python, C/C++, **React Native**, Spring Boot 3, Node.js

AI & Automation: Agentic AI, RAG Pipelines, Google ADK, LLMs, NLP, FAISS Vector DB

Databases & Cloud: PostgreSQL, MySQL, Supabase, **Firebase**, Redis, Kafka, Docker, Kubernetes, AWS (DVA-C02)

Core Concepts: REST APIs, Client-Server Architecture, Distributed Systems, Multi-threading, Agile, CI/CD, Git

PROFESSIONAL EXPERIENCE

EaselearnAI

Full Stack Developer Internship (AI & Mobile)

June 2025 – July 2025

- Engineered a **React Native** full-stack mobile app (5k+ users); designed responsive UI components and architected a high-performance WebView to ensure seamless cross-platform user experiences.
- Developed custom **Google OAuth** flows with secure session management, resolving critical authentication failures and integrating frontend components with backend security protocols.
- Developed an **Agentic AI HR Assistant** in Go; automated leave/attendance workflows via Telegram Bot API, integrating backend services with third-party messaging APIs for real-time orchestration.

TECHNICAL PROJECTS

Real-Time Developer Activity Analytics Pipeline

Python, Apache Kafka, Docker, AWS Lambda, AWS S3

- Architected a data pipeline to ingest streaming event payloads from the live **GitHub Public API** into a locally containerized Kafka cluster while managing API token rate-limiting and pagination handles.
- Engineered an **AWS Lambda** parser to filter event patterns and push lightweight aggregated developer metric summaries into **AWS S3** optimized storage within standard free tier limits.

Distributed Fault-Tolerant Compute Cluster

Python, Distributed Systems, TCP/IP, OS Kernels

- Orchestrated a resilient Master-Slave cluster via raw **TCP sockets**, managing low-level OS resources to achieve linear scalability and deterministic communication protocols for distributed nodes.
- Engineered dynamic load-balancing algorithms to optimize CPU throughput and implemented multi-threaded networking configurations with heartbeat signals to eliminate single points of failure.

Legal Research Assistant: Hybrid AI Automation System

Python, RAG, FAISS Vector DB, LLMs, NLP

- Engineered a robust **RAG pipeline** utilizing Vector Embeddings and Semantic Search to retrieve legal data with 95% accuracy, reducing manual research time from hours to seconds.
- Built a Hybrid system combining LLMs with a forward-chaining inference engine to provide transparent advice, aligning with **Agentic AI** trends and structured **REST API** routing.

PayPipe: High-Availability Payment Ecosystem

Java 21, Spring Boot 3, Kafka, Redis, PostgreSQL

- Architected **event-driven microservices** using Apache Kafka to ensure unified, trusted data flow and transactional consistency across distributed clusters in high-concurrency environments.
- Eliminated race conditions via **Redis Distributed Locks** and enforced 90%+ code coverage using JUnit 5 in a TDD environment to ensure production-grade reliability.
- Optimized memory management and processing throughput using **Java Streams and JPA**, ensuring backend service performance for data-intensive financial workloads.

ACHIEVEMENTS & LEADERSHIP

Caterpillar Hackathon Finalist (GitHub)

Project: Automated CAT Inspect (Python, NLP, OpenCV)

Built a voice-to-text tool to extract machine conditions, improving diagnostic accuracy via AI-powered experience orchestration.

Certifications: Java Full Stack Development | AWS Certified Developer (DVA-C02) | **IEEE-CS Core Committee**